$_3P_2$ $\binom{10}{5}$ 0/26 M. ①

Review.

1. Flip a coin and roll a die.

# of results. $2 \times 6 = 12$.

2. 4 Red balls, 3 Black balls, 2 White Balls.

permutation (if they are different). # of ways to line up. $\boxed{9!}$

If 4 Red Balls are the same. # of ways to line up. $\frac{9!}{4!}$

Probability - describes the proportion of times an event occurs in long-run.

Rule: If S contains N equally likely outcomes, then

1> probability of any outcome is $\frac{1}{N}$.

2> probability of event A is $P(A) = \frac{\#A}{\#S}$

Example: Roll a die 2 times

a> S = { 11 12 $\boxed{13}$ 14 15 16
21 22 23 $\boxed{24}$ 25 26
$\boxed{31}$ 32 33 34 $\boxed{35}$ 36
41 $\boxed{42}$ 43 44 45 $\boxed{46}$
51 52 $\boxed{53}$ 54 55 56
61 62 63 $\boxed{64}$ 65 66 }

all equally likely happen

b> A = {16, 23, 32, 61}

$P(A) = \frac{\#A}{\#S} = \frac{4}{36} = \frac{1}{9}$.

c). B = absolute difference is 2.

$$P(B) = \frac{\#B}{\#S} = \frac{8}{36} = \frac{2}{9}.$$

Example.: # of ways of being dealt <u>10 cards</u>

♡ 1 – 13

♢ 1 – 13

♣ 1 – 13

♤ 1 – 13

No jokers.

$\underline{13 \times 4 = 52}$

a). Total # of ways. $\binom{52}{10}$

b). Probability to have 5 hearts.

A = { exact 5 hearts }.

$$\#A = \binom{13}{5}\binom{39}{5}$$

$$P(A) = \frac{\#A}{\#S} = \frac{\binom{13}{5}\binom{39}{5}}{\binom{52}{10}} = 0.136$$

c). probability to have more than 1 heart.

B = { more than 1 heart }

$$\#B = \binom{13}{2}\binom{39}{8} + \binom{13}{3}\binom{39}{7} + \cdots \binom{13}{10}$$

$$B = S - B'$$

$$\#B = \#S - \#B'$$

$$\#B' = \binom{39}{10} + \binom{13}{1}\binom{39}{9}$$

$$P(B) = \frac{\#S - \#B'}{\#S} = 1 - \frac{\#B'}{\#S} = 1 - P(B')$$

$$= 1 - \frac{\binom{39}{10} + \binom{13}{1}\binom{39}{9}}{\binom{52}{10}} =$$

Additive Rules

① . $P(S) = 1.$ $\qquad P(\phi) = 0$

②. For any event A

$$0 \le P(A) \le 1$$

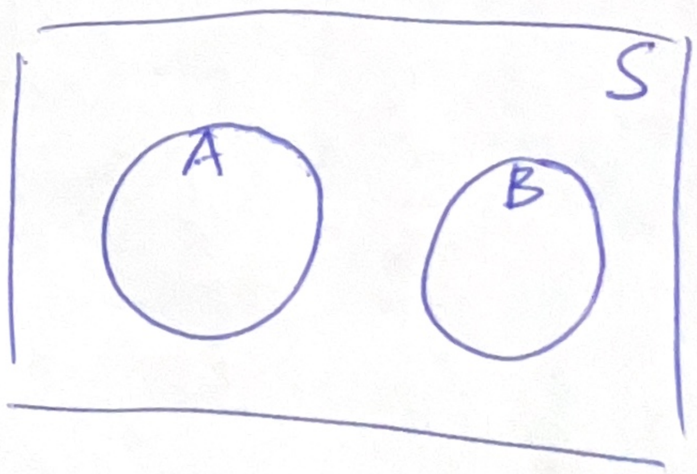
③ Complement rule.

$$P(A') = 1 - P(A) \qquad \text{or} \qquad P(A) = 1 - P(A')$$

④ Additive Rule for m.e. events.

If A and B are m.e. then

$$P(A \cup B) = P(A) + P(B)$$

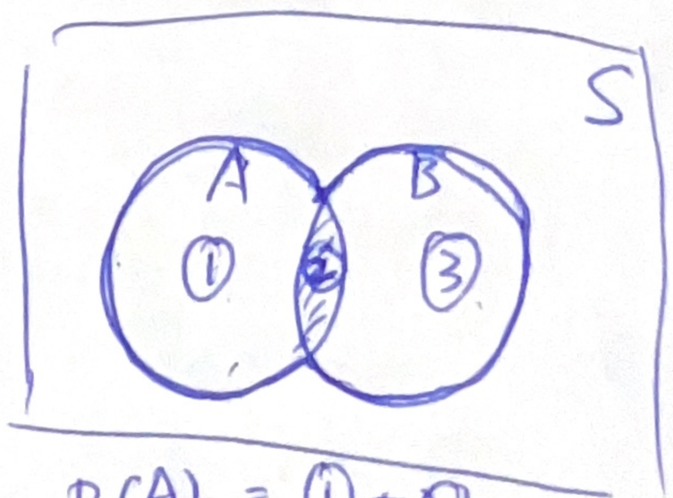


⑤ General Additive Rule.

For any events A and B.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

⑥. If $A \subseteq B$, then $P(A) \le P(B)$



$P(A) = ① + ②$

$P(B) = ② + ③$

$P(A \cap B) = ②$

$P(A \cup B) = ① + ② + ③$